

Aprameya Satish

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Active Secret Clearance

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EDUCATION

- **Georgia Institute of Technology** Atlanta, GA
Ph.D in Mechanical Engineering; GPA: 4.0 Aug 2015 - Dec 2020
- **Georgia Institute of Technology** Atlanta, GA
MS in Electrical and Computer Engineering; GPA: 4.0 Aug 2017 - Dec 2020
- **BMS College of Engineering** Bangalore, India
Bachelor of Mechanical Engineering Aug 2009 - June 2013

SKILLS

- **Languages:** Python, Matlab, C/C++
- **Domain Knowledge:** Applied Machine learning for Acoustics, RADAR, SONAR, and health; Statistical Signal Processing; Spatial Array Processing; Acoustics and Infrasound; Elastic Wave Propagation

EXPERIENCE

- **Georgia Tech Research Institute** Smyrna, GA
Senior Research Engineer Dec 2020 - Current
 - **Vibroacoustic based Pipeline Leak Detection:** Technical lead responsible for experiment design, data management, and statistical DSP and ML algorithm development for pipeline leak detection using vibroacoustic datastreams.
 - **Cognitive Radar Modeling:** Technical lead for ML algorithm development for new cognitive radar program at GTRI, sponsored by Missile and Space Intelligence Center (MSIC)
 - **Soldier fitness and health monitoring:** Key contributor in GTRI's signal processing and machine learning work in a multi-million dollar, cross-collaborative effort with Georgia Tech on ONR sponsored military health monitoring and prediction project, resulting in multiple publications and a patent.
 - **Infrasound source localization and classification:** Developed signal processing pipeline that uses measurements from infrasound sensor arrays to localize infrasound targets of interest. Built ML models for infrasound signature classification, and developed synthetic infrasound signature modeling software suite.
 - **SAR Automatic Target Recognition:** Key contributor in the signal processing and deep-learning based target detection efforts for AFLCMC sponsored automatic target detection and recognition using SAR data.
 - **Computer-Vision based underwater target detection:** Technical lead on internal program to build machine learning algorithms for underwater landmark detection from SONAR data for autonomous vehicle localization.
 - **UAV tracking using acoustic beamforming:** Headed algorithm development and for outdoor UAV tracking using spherical array acoustic beamforming for DARPA program.
 - **Infrasound and Acoustic exposure analysis:** Working with WRAIR (US Army), NSWC, Raytheon, on mitigating sound and infrasound exposure to soldiers during weapons and blast training.
 - **Maritime Infrasound Monitoring System:** Coordinated the full wireless communications and signal processing pipeline for a realtime infrasound beamforming dashboard for the ONR.
 - **Internal Research Management:** Secured and managed \$210k of internal funding to develop new technologies in underwater navigation and threat detection using acoustics, machine learning for health.

Graduate Research Assistant

Jan 2018 - Jul 2018

- **Auditory UAV based source localization:** Worked on acoustic source localization in low and very low SNR environments using ground-mounted and Micro-Aerial Vehicle (MAV/drone) mounted microphone arrays.
- **Monochannel audio source separation:** Implemented a statistical machine learning framework for acoustic source and noise separation based on latent variable decompositions of audio recordings that works with limited training data.
- **Jet Noise analysis using NMF:** Applied non-negative matrix factorization to study jet engine mode-switching - a combustion instability.
- **SpaceX** Hawthorne, CA
Propulsion Dynamics Intern May 2016 - Aug 2016
 - **Merlin Engine Dynamics Analysis:**
 - Analysed engine dynamics from pressure, accelerometer, and strain gauge data for the Merlin Stage I and vacuum engines to qualify engine performance.
 - Performed statistical analyses on combustion chamber pressure and peripheral sensor measurements to define nominal operating parameters for Stage 1 and vacuum engines.
 - Automated the process of evaluating engine fatigue using rainflow analyses on strain gauge data.

PHD THESIS

A. Satish, *Analysis and Design of Passive Underwater Acoustic Identification Tags*. PhD thesis, Georgia Institute of Technology, 2020

REFEREED JOURNAL PAPERS

1. D. J. Lin, **Satish, A.**, K. Richardson, S. An, C. Yaldiz, M. Buller, K. Driver, E. Atkinson, T. Mesite, C. King, O. Inan, and A. Medda, "Predicting soldier performance on structured military training marches with wearable accelerometer and physiological data," *IEEE Sensors*, 2023
2. C. Yaldiz, M. Buller, K. Richardson, S. An, D. J. Lin, **Satish, A.**, K. Driver, E. Atkinson, T. Mesite, C. King, M. Bursey, M. Galer, M. M. Stafford, M. Sawka, A. Medda, and O. Inan, "Early prediction of impending exertional heat stroke with wearable multimodal sensing and anomaly detection," *IEEE Journal of Biomedical and Health Informatics*, 2023
3. **Satish, Aprameya** and K. G. Sabra, "Passive underwater acoustic identification tags using multi-layered shells," *The Journal of the Acoustical Society of America*, vol. 149, no. 5, pp. 3387–3405, 2021
4. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Omnidirectional passive acoustic identification tags for underwater navigation," *The Journal of the Acoustical Society of America*, vol. 147, no. 6, pp. EL517–EL522, 2020
5. **Satish, Aprameya**, B. Nichols, D. Trivett, and K. G. Sabra, "Passive underwater acoustic tags using layered media," *The Journal of the Acoustical Society of America*, vol. 145, no. 1, pp. EL84–EL89, 2019

CONFERENCE PROCEEDINGS

1. **Satish, Aprameya**, E. Laubie, J. Casey, and T. Selee, "Automatic target chip extraction from sar images using cnns and cfar," in *Automatic Target Recognition Working Group Meeting*, National Geospatial Intelligence Agency, 2023
2. **Satish, Aprameya** and A. Medda, "Acoustic uav detection using spherical array beamforming," in *2022 Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2022
3. **Satish, Aprameya**, A. Karon, and A. Medda, "Predicting supersonic jet mode-switching using non-negative matrix factorization," *The Journal of the Acoustical Society of America*, vol. 150, no. 4, pp. A177–A177, 2021
4. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Passive underwater acoustic tags with curved symmetry for navigation and information encoding," *The Journal of the Acoustical Society of America*, vol. 145, no. 3, pp. 1937–1937, 2019
5. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Experimental evaluation of passive underwater acoustic markers," *The Journal of the Acoustical Society of America*, vol. 146, no. 4, pp. 3019–3019, 2019
6. **Satish, Aprameya** and A. Medda, "Time-frequency beamforming adaptation for micro aerial vehicle (mav) noise suppression and source localization," in *2018 52nd Asilomar Conference on Signals, Systems, and Computers*, pp. 884–888, IEEE, 2018
7. **Satish, Aprameya**, B. Nichols, D. Trivett, and K. G. Sabra, "Passive underwater acoustic markers for navigation and information encoding for high frequency sound navigation and ranging (sonar) devices," *The Journal of the Acoustical Society of America*, vol. 142, no. 4, pp. 2731–2731, 2017
8. A. Spexet, N. Breen, **Satish, Aprameya**, and D. Alvord, "Intelligent summarization of aviation data for maintenance," in *2024 IEEE Aerospace Conference*, pp. 1–10, IEEE, 2024
9. W. N. Newcomb, S. Reddy, **Satish, Aprameya**, and A. Medda, "Time-frequency spectral optimization of underwater sonar acoustic returns for the purpose of characterization using spectral descriptors," *The Journal of the Acoustical Society of America*, vol. 156, no. 4_Supplement, pp. A25–A26, 2024
10. S. Das, **Satish, Aprameya**, N. Breen, M. Fitzpatrick, and A. Medda, "Development of synthetic infrasound signatures for target classification using window of opportunity observations," *The Journal of the Acoustical Society of America*, 2025

PROFESSIONAL SERVICE

- Treasurer, IEEE Atlanta Chapter (*Feb 2022 - Feb 2025*)
- Technical Co-Chair, IEEE SoutheastCon 2024 (*Aug 2022 - present*)
- Reviewer, Measurement (*Dec 2020 - present*)
- Reviewer, Acoustical Society of America (*Dec 2020 - present*)
- Grant Committee Member, Melton Foundation (*Dec 2020 - present*)
- Grad Gala Operations and Logistics Chair, GT Graduate Student Association (*Mar 2020*)
- Treasurer, Acoustical Society of America GT Student Chapter (*Dec 2017*)
- President's Undergraduate Research Award (PURA) Grant Committee Technical Reviewer, (*Jun 2016, Jun 2020*)
- Co-Organizer and Compliance Officer, TEDxBangalore (*Aug 2014*)
- Senior Fellow, Melton Foundation (*Jun 2013 - present*)

RECOGNITION

- Outstanding Service Award, IEEE Atlanta Section, (*Oct 2024*)
- Outstanding Service Award, GT Graduate Student Association, (*Mar 2020*)
- GTRI Best Internally Funded project (IRAD) nominee, (*Jun 2018*)
- Mercedes-Benz RD Star Performer nominee, (*Jun 2014*)